

Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an adherent point of A if every open neighbourhood U of x satisfies

$$U \cap A \neq \emptyset.$$

In the special case where X is a metric space with metric d , this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \emptyset.$$

Definition 2. Let X be a topological space and $A \subseteq X$. The set A is called dense in X if every nonempty open set $U \subseteq X$ satisfies

$$U \cap A \neq \emptyset.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. Let X be a topological space and $A \subseteq X$. The set A is called nowhere dense in X if

$$\text{int}(\overline{A}) = \emptyset.$$

Definition 4. A topological space X is separable if: $\exists D \subseteq X : D$ countable and dense.

Baire Category Theorem

Theorem 1 (Baire Category Theorem). Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X . Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \quad r_1 < 1.$$

Inductively, having chosen y_n, r_n , use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \quad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \rightarrow 0$. Thus (y_n) is Cauchy, so by completeness $y_n \rightarrow y \in X$.

For each fixed n , all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \emptyset,$$

so the intersection is dense. □

Remark 1. From the definitions it follows that a set $B \subseteq X$ satisfies $\text{int}(B) = \emptyset \iff X \setminus B$ is dense in X . In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X .

Corollary 1. Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X . Then

$$\text{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \emptyset.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. Let (X_k, d_k) , $k = 1, \dots, n$, be metric spaces and set

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric if convergence in (X, d) is coordinatewise,

$$x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)}) \rightarrow x = (x_1, \dots, x_n)$$

in X if and only if for each $k = 1, \dots, n$,

$$x_k^{(m)} \rightarrow x_k \quad \text{in } X_k.$$

Theorem 2. Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k .

Proof. First, if $E_k \subset X_k$ is closed for each k , then

$$E_1 \times \cdots \times E_n$$

is closed in X . Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \rightarrow x$ in X , then by coordinatewise convergence, $x_k^{(m)} \rightarrow x_k$ in X_k . Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X , since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^n V^{(k)}$$

is open in X , and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \dots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \quad k = 1, \dots, n.$$

Then $x_k^{(m)} \rightarrow x_k$ for each k , hence $x^{(m)} \rightarrow x$ in X . Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. $\square$

Theorem 3 (Heine–Borel Theorem). *For a subspace $E \subseteq \mathbb{R}^n$, the following are equivalent:*

- (i) E is compact;
- (ii) every sequence in E has a convergent subsequence with limit in E ;
- (iii) E is closed and bounded.

Proof. By the general compactness theorem for metric spaces, for any metric space the following are equivalent: compactness, sequential compactness, and completeness together with total boundedness.

Now let $E \subseteq \mathbb{R}^n$.

If E is closed and bounded, then E is complete (since a subspace of $\mathbb{R}^n$ is complete if and only if it is closed) and totally bounded (since a subset of $\mathbb{R}^n$ is totally bounded if and only if it is bounded). Hence E is compact, and therefore every sequence in E has a convergent subsequence.

Conversely, if E is compact, then by the same general theorem E is complete and totally bounded. Since E is a subspace of $\mathbb{R}^n$, completeness implies that E is closed, and total boundedness implies that E is bounded. Thus E is closed and bounded.

Therefore (i), (ii), and (iii) are equivalent. $\square$

Topologies

Definition 6. Let X be a set. A topology on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,

(ii) if $\{U_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$, then

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T},$$

(iii) if $U_1, \dots, U_n \in \mathcal{T}$, then

$$\bigcap_{i=1}^n U_i \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is called a topological space. The sets $U \in \mathcal{T}$ are called open sets. A subset $F \subseteq X$ is called closed if its complement is open.

Separation Axioms

Definition 7 (T_0 space). A topological space $(X, \mathcal{T})$ is called a T_0 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$x \in U, y \notin U \quad \text{or} \quad y \in U, x \notin U.$$

Definition 8 (T_1 space). A topological space $(X, \mathcal{T})$ is called a T_1 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$y \in U \quad \text{and} \quad x \notin U.$$

Equivalently, $(X, \mathcal{T})$ is T_1 if and only if every singleton set $\{x\}$ is closed in X .

Definition 9 (T_2 space / Hausdorff). A topological space $(X, \mathcal{T})$ is called a T_2 -space, or Hausdorff, if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad y \in V, \quad U \cap V = \emptyset.$$

Definition 10 (Regular space). A topological space $(X, \mathcal{T})$ is called regular if for every closed set $F \subseteq X$ and every point $x \in X \setminus F$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad F \subseteq V, \quad U \cap V = \emptyset.$$

Definition 11 (T_3 space). A topological space $(X, \mathcal{T})$ is called a T_3 -space if it is both T_1 and regular.

Definition 12 (Normal space). A topological space $(X, \mathcal{T})$ is called normal if for any two disjoint closed sets $F, G \subseteq X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$F \subseteq U, \quad G \subseteq V, \quad U \cap V = \emptyset.$$

Definition 13 (T_4 space). A topological space $(X, \mathcal{T})$ is called a T_4 -space if it is both T_1 and normal.

Theorem 4. *Every compact Hausdorff space is normal. In particular, every compact Hausdorff space is a T_4 -space.*

Proof. Let $F, G \subseteq X$ be disjoint closed sets. Since X is Hausdorff, for each $x \in F$ and $y \in G$ there exist disjoint open sets $U_{x,y}$ and $V_{x,y}$ such that

$$x \in U_{x,y}, \quad y \in V_{x,y}, \quad U_{x,y} \cap V_{x,y} = \emptyset.$$

For fixed $x \in F$, the sets $\{V_{x,y}\}_{y \in G}$ form an open cover of G . Since G is closed in the compact space X , it is compact, so we may choose a finite subcover

$$G \subseteq V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Let

$$U_x := U_{x,y_1} \cap \cdots \cap U_{x,y_n}, \quad V_x := V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Then $x \in U_x$, $G \subseteq V_x$, and $U_x \cap V_x = \emptyset$.

The family $\{U_x\}_{x \in F}$ is an open cover of F . Since F is compact, choose a finite subcover

$$F \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Define

$$U := U_{x_1} \cup \cdots \cup U_{x_m}, \quad V := V_{x_1} \cap \cdots \cap V_{x_m}.$$

Then U and V are open, $F \subseteq U$, $G \subseteq V$, and $U \cap V = \emptyset$. Hence X is normal. $\square$

Basis & Subbasis

Definition 14. *Let X be a set. A collection $\mathcal{B}$ of subsets of X is called a basis for a topology on X if:*

- (i) *For every $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$.*
- (ii) *If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that*

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 5 (Topology generated by a basis). *If $\mathcal{B}$ satisfies the two conditions above, then there exists a unique topology $\mathcal{T}$ on X whose open sets are precisely the unions of elements of $\mathcal{B}$. That is,*

$$\mathcal{T} = \left\{ \bigcup_{\alpha \in I} B_\alpha : B_\alpha \in \mathcal{B}, \text{ } I \text{ arbitrary} \right\}.$$

Definition 15. *Let X be a set. A collection $\mathcal{S}$ of subsets of X is called a subbasis for a topology on X if the union of all sets in $\mathcal{S}$ equals X ,*

$$X = \bigcup_{S \in \mathcal{S}} S.$$

The topology generated by $\mathcal{S}$ is obtained by first forming the basis consisting of all finite intersections of elements of $\mathcal{S}$,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n : S_i \in \mathcal{S}, \text{ } n \in \mathbb{N}\},$$

and then taking arbitrary unions of these sets.

The topology generated by the subbasis $\mathcal{S}$ consists of all arbitrary unions of elements of this basis. That is, a set $U \subseteq X$ is open if and only if it can be written as

$$U = \bigcup_{\alpha \in I} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha,i} \right), \quad S_{\alpha,i} \in \mathcal{S}, \quad n_\alpha \in \mathbb{N}.$$